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[Python] Events x Tracking Synchronization

Sync your Impect, Opta, Statsbomb, Wyscout Events with SkillCorner Tracking Data v3

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In this tutorial, you will learn how to synchronize event data from sources like Impect, Opta, StatsBomb, or Wyscout with SkillCorner Tracking Data v3 using Python.

If you want to know the methodology used to synchronise events and tracking data, please read this article.

If you are still working with SkillCorner Tracking Data v2 you can should use the code available in this article.

> **Need some Help?**
> Send us an email at *support@skillcorner.com.*

The output

The output of the synchronisation code consists in the follow files:

- a console output reflecting **matching performance**
- a list of json named **freeze_frame_format.json**
- a csv file **report_by_event.csv**
- a csv file **report_by_event_type.csv**

```
=================== MATCH INFO ===================
- match_name: 20220508_Paris_Troyes AC
- match_id: 127443
- event_provider: wyscout
==================================================
Data loading took 1.52 seconds.
```

wyscout_event_type	nb_events	is_matched	%_is_matched
acceleration	11	8	72.7
clearance	8	6	75
corner	11	6	54.5
duel	342	280	81.9
free_kick	30	15	50
goal_kick	17	1	5.9
goalkeeper_exit	8	3	37.5
infraction	31	24	77.4
interception	46	39	84.8
offside	3	3	100
pass	940	880	93.6
penalty	2	2	100
shot	24	20	83.3
shot_against	10	10	100
throw_in	26	16	61.5
touch	132	129	97.7

```
Event synchronization process took 2.83 seconds.
```

Console output reflecting matching performance

freeze_frame_format.json, a list of dictionaries containing enriched tracking data by event frame. Example of one element of the list:

```
{
    "frame": 35261,
    "timestamp": "00:53:28.10",
    "period": 2,
    "ball_data": {
        "x": -25.15,
        "y": -24.01,
        "z": 0.17,
        "is_detected": true
    },
    "player_data": [
        {
            "x": -46.05,
            "y": -2.62,
            "player_id": 11898,
            "is_detected": false,
            "speed_norm": 1.47,
            "vx": -1.45,
```

Key	Description
frame	Frame associated with the event.
timestamp	Match timestamp.
period	Integer that represents the period associated with the freez frame. First half corresponds to 1, Second Half to 2, First Extra-Tim corresponds to 3 and Second Extra-Time corresponds to 4.
ball_data	Ball tracking data.
possession	Dictionary with team and player possession.
image_corners_projection	Coordinates of the projection of the camera field of view on pitch plane.
player_data	Players tracking data. Speeds and accelerations are included. They can be censored when deemed unrealistic.
{provider}_event_id	Event Provider Event ID
{event_provider}_event_type	Event Provider Event Type associated with the event.
player_id	SkillCorner Player ID associated with the event.
{event_provider}_player_id	Event Provider Player ID associated with the event.
team_id	SkillCorner Team ID associated with the Event.
{event_provider}_team_id	Event Provider Team ID associated with the Event.
is_matched	True if the event is matched with the Tracking. Else False.
is_player_detected	True if the player associated with the event is detected by SkillCorner in the field of view of the camera. Else False.

projected_{event_provider}_event_x	x coordinate of the event according to the Event Provider projected in SkillCorner system of coordinates.
projected_{event_provider}_event_y	y coordinate of the event according to the Event Provider projected in SkillCorner system of coordinates.

> **Speeds and accelerations**
>
> As for previous versions of the code, Speeds and Accelerations are delivered.
> They can be censored when the values are unrealistic.
>
> At frame N,
> - the player speed is simply the difference between its position at frame N+1 and frame N-1 divided by 0.2.
> - the player acceleration is the difference between its speed at frame N+4 and frame N-4 divided by 0.8.

report_by_event.csv, a CSV file where each row corresponds to an event. Columns:

Column	Description
{event_provider}_event_id	Event Provider Event ID.
frame	Frame associated with the event.
player_id	SkillCorner Player ID associated with the event.
{event_provider}_player_id	Event Provider Player ID associated with the event.
player_name	SkillCorner Player Name
player_role	SkillCorner Player Name (e.g. GK, LB).
player_number	Jersey Number of the Player

starting	True if the player is starting the match. Else False.
team_id	SkillCorner Team ID of the player.
{event_provider}_team_id	Event Provider Team ID of the player.
team_type	"home_team" if the player is team is the home team. "away_team" if the player is team is the away team.
team_name	SkillCorner name of the Team.
is_matched	True if the event is matched with the Tracking. Else False.
is_player_detected	True if the player is detected to generated his location.
has_{event_provider}_player_id_attached	True if the Event Provider Attached a Player ID to the Ev Else False.
frame_tracking_data_available	True if SkillCorner Deliver Tracking Data for this particu Frame. Else False.
is_matched_applicable	True if matching the event is relevant. Else False (e.g. substitutions).

report_by_event_type.csv, a CSV file grouping events by type with performance metrics related to is_matched. Columns include:

Column	Description
{event_provider}_event_type	Type of the event.
nb_events	Number of occurrences of the event.
is_matched	Number of events that are matched with the tracking data.
%_is_matched	% of matched events.
is_matched_is_player_detected	Number of events matched where the player w detected.
is_not_matched	Number of events that are not matched with tl tracking data.
is_not_matched_is_player_detected	Number of events not matched where the play was detected.
is_not_matched_has_{event_provider}_player_id_attached	Number of events not matched where the Ever Provider attached a player ID to the event.
is_not_matched_frame_tracking_data_available	Number of events not matched but where trac data is delivered at the frame of the event.
is_matched_applicable	True if matching the event is relevant. Else False (e.g. substitutions).

Prerequisites

- Python 3.8 or above installed.
- SkillCorner Python Client. If you don't have it, refer to this article.
- Input files: The synchronisation code requires several files to run. You can either store these files prior to the synchronisation or gather them on the fly.
  - SkillCorner Match Data
  - SkillCorner Tracking Data v3
  - Event Provider Event File
  - Event Provider Match Data

SkillCorner Match Data

It is the json that contains the lineups for the match as well as some additional information.

You can retrieve it

- Directly via the API endpoint /match/{match_id}.
  Then, you need to save it.
- Via the SkillCorner Python Client save_match() method.
  Example: `client.save_match(match_id=12345, file_path("/path/where/to/save/12345_match_data.json")`

> **Wyscout Specific**
> If you want to sync the tracking data with Wyscout Events you need to retrieve the SkillCorner Data with mapping between SkillCorner IDs and Wyscout IDs.
>
> To do so
> - request the endpoint /match/{match_id} adding the query parameter matching=wyscout
> - or if you use the Python client, add the parameter to the method:
> `client.save_match(match_id=12345, file_path("/path/where/to/save/12345_match_data.json", params={'matching':'wyscout'})`

SkillCorner Tracking Data v3

It is the Tracking Data V3 from SkillCorner in the jsonl format.

- Directly via the API endpoint /match/{match_id}/tracking?data_version=3.
- Via the SkillCorner Python Client save_match_tracking_data() method.
  Example: `client.save_match_tracking_data(match_id=12345, file_path("/path/where/to/save/12345_tracking_data.jsonl", params={'data_version':3})`

Event Provider Event File

This is your Event Provider file that contains the events:

- For Wyscout, it corresponds to the json file from the Wyscout api v3 via the endpoint /matches/{wyscout_match_id}/events
- For Opta, it corresponds to
  - either the opta f24 xml file named like f24-x-xxxx-xxxxxxx-eventdetails.xml,
  - or the json file that contains the event data.
- For Statsbomb, it corresponds to the json file containing the events.
- For Impect, it corresponds to the json file containing the events.

Event Provider Match Data

This is your Event Provider file that contains the lineups.

- For Wyscout, you don't need it.
- For Opta it corresponds to
  - either the opta f7 xml file named like srml-x-xxxx-fxxxxxxx-matchresults.xml,
  - or the json file that contains the match data.
- For Statsbomb, it corresponds to json file that contains the lineups.
- For Impect, it corresponds to the json file that contains the lineups.

> **Opta specific**
> You need to be consistent in the format of the Opta file you decide to use.
> - If you use the XML files for the Events, you need to use the XML file for Opta Match Data.
> - Conversely if you use the json file for the Events you need to use the json file for the Opta Match Data.

Step by Step instructions

1/ Download the synchronisation code

Download the Python script from the following link: Event Synchronisation Code

It is a zip file you need to unzip.

2/ Install required libraries

From the root directory you have unzipped, in your terminal, run the following command .

```
pip install -r requirements.txt
```

3/ Run the Code

From your terminal

In addition to the previous input you need to have an *Output Directory* where you want the files, report_by_event_type.csv, freeze_frame_format.json and report_by_event.csv, to be saved.

1. Ensure you are in the root of folder *skillcorner-event-synchronization*.
2. Run the event-synchronization script:

For Impect

```
python -m tools.run_impect.py
    --match_data_path /path/to/skc_match_data.json \
    --tracking_data_path /path/to/skc_tracking_data.json \
    --impect_events_path /path/to/impect_events.json \
    --impect_match_data_path /path/to/impect_match_data.json \
    --save_outputs_dir /path/to/output_directory
```

For Statsbomb

```
python -m tools.run_statsbomb.py \
    --match_data_path /path/to/skc_match_data.json \
    --tracking_data_path /path/to/skc_tracking_data.json \
    --statsbomb_events_path /path/to/statsbomb_events.json \
    --statsbomb_match_data_path /path/to/statsbomb_lineup.json \
    --save_outputs_dir /path/to/output_directory
```

For Opta

```
python -m tools.run_opta.py \
    --match_data_path /path/to/skc_match_data.json \
    --tracking_data_path /path/to/skc_tracking_data.json \
    --opta_events_path /path/to/opta_events.json (or .xml) \
    --opta_match_data_path /path/to/opta_match_data.json (or .xml) \
    --save_outputs_dir /path/to/output_directory
```

For Wyscout

```
python -m tools.run_wyscout.py \
    --match_data_path /path/to/skc_match_data.json \
    --tracking_data_path /path/to/skc_tracking_data.json \
    --wyscout_events_path /path/to/wyscout_events.json \
    --save_outputs_dir /path/to/output_directory
```

From a notebook

You can find notebooks for every provider under

`skillcorner-event-synchronization/tools/notebook_example`

Troubleshooting

If you encounter errors, check the following:

- Ensure the file paths are correct.
  If using the terminal commands, the directory from which you are executing the code should be *skillcorner-event-synchronization.*
- Verify the data formats match the expected input.

If you are still blocked contact our support at *support@skillcorner.com*.

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